NEPSE Forecasting Study — Recovered Floorsheet Days and Their Effect

The three original floorsheet archives were missing or near-empty on 280 of 1,858 sessions. Eighty-eight recovered SQL day files have now been merged in. This report records what they contained, what is still missing, and what the recovery does to every headline result. Three runs are compared on an identical, untouched final test: the original archives with the thin sessions kept, the original archives with them excluded, and the supplemented archives with only the still-missing sessions excluded.

Summary

The recovery works. Excluding the thin sessions had cost accuracy at 3, 5 and 10 sessions; feeding the real data back recovers almost all of it while keeping the clean design. At 10 sessions accuracy goes 55.3% → 50.5% → 54.8% across the three runs, and the ranker's rank-IC returns to where it was with the full original sample.

The 1-session result is stable across all three. 59.0%, 58.6% and 58.5% against a 55.3% majority baseline, each with a bootstrap probability below 0.001 of being a baseline-level result. Its calibration is best in the recovered run (expected calibration error 2.50% against 3.23% and 3.15%), and the 70%-confidence operating point covers more of the test than either earlier run: 301 predictions at 80.4% accuracy, against 243 at 81.1% and 180 at 83.9%.

The recovered files are clean. 3,687,558 trades parsed with no unparsed lines, no rows dropped for nulls, non-positive values or bad broker codes, no duplicate transaction numbers, no amount mismatches, no serial-number/date mismatches and no OHLC violations. They are a strict superset of what the originals held for those dates: no bar had to be kept from the old data, and not one replaced bar had fewer trades than before.

One block is still missing: 2022-07-28 to 2023-03-30, 154 sessions. With its 19-session buffer and the isolated thin days, 173 sessions remain excluded from training, down from 317.

What the recovered files contained

	Before	After
Trades in the floorsheet	91,009,140	94,671,327
Daily symbol-session bars	405,370	425,907
Trades on the 97 recovered sessions	25,371	3,687,558
Sessions below 50% coverage	280	183
Panel floorsheet coverage	88.5%	98.8%

The merge replaced 2,136 existing bars and added 20,537 new ones across 97 sessions. 1,618 broker fields carried the alphabetic dealer code `D01`, mapped to the same identifier the original ingest used.

Coverage, block by block

Block	Sessions	Status	Detail
2022-02-10 → 2022-02-28	13	**recovered**	All 13 sessions supplied in full
2022-03-31 → 2022-07-27	84	**recovered**	Supplied except 10 Fridays (see below)
2022-07-28 → 2023-03-30	154	still missing	0–21% coverage; the one remaining structural block
2024-03-17 → 2024-03-31	10	**recovered**	All 10 sessions, 550,999 trades

Still worth chasing. The 154-session block above; ten Fridays inside the recovered stretch — 2022-05-20, 05-27, 06-03, 06-10, 06-17, 06-24, 07-01, 07-08, 07-15 and 07-22; and the isolated days 2018-11-11, 2018-12-31, 2019-01-14, 2019-01-22, 2019-03-09, 2019-09-12, 2020-07-26, 2020-08-26, 2020-08-30, 2021-03-08, 2021-10-05, 2025-04-01, 2025-07-22, 2026-03-02, 2026-08-28, 2026-09-01, 2026-09-04, 2026-09-08 and 2026-10-23.

The exclusion rule, applied to what is still missing

The rule is unchanged: a session is thin when fewer than 50% of its eligible stocks appear in the floorsheet; runs of at least 5 consecutive thin sessions are dropped along with the 19 sessions that follow, so no rolling or sequence window reads a thin session. Because the data is now there, the rule simply has less to remove.

	Original archives	Supplemented
Sessions excluded	317	173
Structural blocks found	3	1
Eligible stock-sessions	305,474	333,786
Sessions in the panel	1,538	1,682
Floorsheet coverage after exclusion	99.2%	98.8%

Training recovers 28,312 stock-days against the excluded run. The walk-forward design keeps eight folds, now with 109-session validation blocks instead of 91. The leakage and split audit passes all ten checks: rebuilding every feature from sources truncated at 2024-10-28 reproduces the stored values exactly (max |difference| 0.0, no missing-data-pattern mismatches, 60 features × 278,781 rows).

The final test window is identical in all three runs — 2025-10-12 to 2026-11-05, 250 sessions, and exactly 60,526 scored rows at 1 session, 61,681 at 10. Every number below is measured on the same rows.

Final-test accuracy

H	Run	Candidate	Base	Acc	95% interval	AUC	Logloss	ECE
1d	original	weighted	55.3%	59.0%	57.0–60.9%	0.608	0.6720	3.23%
	excluded	weighted	55.3%	58.6%	56.5–60.6%	0.604	0.6738	3.15%
	recovered	weighted	55.3%	58.5%	56.4–60.5%	0.602	0.6730	2.50%
3d	original	weighted	57.3%	59.2%	55.7–62.6%	0.610	0.6732	4.88%
	excluded	weighted	57.3%	57.5%	54.8–60.4%	0.585	0.6795	5.69%
	recovered	weighted	57.3%	58.2%	54.9–61.6%	0.586	0.6767	4.66%
5d	original	weighted	57.3%	58.3%	53.7–62.5%	0.597	0.6801	6.23%
	excluded	stacking + platt	57.3%	57.3%	50.1–64.5%	0.473	0.6882	5.26%
	recovered	weighted	57.3%	57.2%	54.1–60.6%	0.585	0.6803	6.01%
10d	original	weighted	56.8%	55.3%	49.1–60.9%	0.584	0.6863	7.32%
	excluded	weighted	56.8%	50.5%	44.6–56.2%	0.588	0.6923	9.98%
	recovered	weighted	56.8%	54.8%	49.9–59.5%	0.577	0.6854	6.92%

`original` = the three original archives with the thin sessions kept; `excluded` = the same archives with them dropped; `recovered` = the supplemented archives with only the still-missing sessions dropped. Each run scores the candidate its own out-of-fold log-loss selected, locked before the test was touched.

Measured like for like on the `weighted` ensemble in all three: 1 session 59.0% / 58.6% / 58.5%; 3 sessions 59.2% / 57.5% / 58.2%; 5 sessions 58.3% / 55.2% / 57.2%; 10 sessions 55.3% / 50.5% / 54.8%. The 5-session `stacking + platt` row in the excluded run, with an AUC of 0.473, is a selection failure rather than a data effect — the `weighted` ensemble scored 0.578 on the same rows. The recovered run's own selection avoids it.

Significance, calibration and the ranker

Horizon	Run	p (acc ≤ base)	DM vs climatology, Holm p	Rank-IC	IC-IR	Spread
1d	original	< 0.001	3.5e-07	0.284	2.50	1.02%
	excluded	< 0.001	1.16e-07	0.281	2.49	1.02%
	recovered	< 0.001	1.01e-06	0.284	2.53	1.01%
3d	original	0.144	0.00632	0.270	2.12	1.45%

Horizon	Run	p (acc ≤ base)	DM vs climatology, Holm p	Rank-IC	IC-IR	Spread
	excluded	0.418	0.0418	0.259	2.03	1.45%
	recovered	0.304	0.0567	0.272	2.12	1.50%
5d	original	0.317	1.0	0.248	1.67	1.67%
	excluded	0.515	1.0	0.234	1.65	1.61%
	recovered	0.521	0.591	0.250	1.63	1.72%
10d	original	0.685	1.0	0.213	1.35	2.04%
	excluded	0.987	1.0	0.195	1.33	1.91%
	recovered	0.785	1.0	0.214	1.36	2.03%

The ranker is the clearest signal that the recovery worked. Exclusion had pulled mean rank-IC down at every horizon (0.270 → 0.259 at 3 sessions, 0.248 → 0.234 at 5, 0.213 → 0.195 at 10); with the real data back it returns to 0.272, 0.250 and 0.214 — at or slightly above the original full-sample run, on a cleaner panel.

Selective prediction at 70% confidence

Horizon	Run	Coverage	Predictions	Accuracy
1d	original	0.40%	243	81.1%
	excluded	0.30%	180	83.9%
	recovered	0.50%	301	80.4%
3d	original	0.03%	22	100.0%
	excluded	0.00%	2	100.0%
	recovered	0.01%	6	83.3%

The 70% operating point remains a narrow one. At 1 session the recovered run makes 301 calls out of 60,526 rows — half a percent of the test — at 80.4% accuracy. That is the most coverage of the three runs, but it is still 301 predictions, and the claim that belongs in a paper is the selective one with its coverage stated, not a headline accuracy.

Backtest

Long-only, entry at the next session's open, 10 bp slippage a side plus commission and the 7.5% short-term capital-gains tax.

Strategy	Run	Trades	Mean net / trade	95% interval	P(mean ≤ 0)
1 session, p ≥ 0.55	original	1,409	-0.332%	-0.507 to -0.158%	1.000
	excluded	1,371	-0.401%	-0.582 to -0.228%	1.000
	recovered	1,496	-0.443%	-0.586 to -0.292%	1.000
3 sessions, p ≥ 0.60	original	244	0.545%	-0.251 to 1.406%	0.118
	excluded	239	0.446%	-0.343 to 1.501%	0.139
	recovered	223	0.103%	-0.642 to 1.369%	0.425
10 sessions, p ≥ 0.60	original	409	0.521%	-1.005 to 2.210%	0.290
	excluded	960	0.376%	-1.286 to 1.780%	0.345
	recovered	561	0.604%	-0.898 to 1.908%	0.231
1 session, ranker top 10	original	2,392	-0.525%	-0.660 to -0.382%	1.000
	excluded	2,393	-0.516%	-0.657 to -0.381%	1.000
	recovered	2,391	-0.500%	-0.632 to -0.351%	1.000

Unchanged in character across all three runs: no strategy earns a net return distinguishable from zero after costs, and the ranker's top-10 portfolio loses about half a percent a trade whatever the training data. Better features did not become a tradable edge.

What to report

Report the recovered run as the primary result. It has the cleanest provenance of the three: every training row carries real floorsheet data, the one period that does not is excluded by a measured rule, and the exclusion is now small enough that it costs almost nothing. The headline is 58.5% at 1 session against a 55.3% baseline on 60,526 held-out rows, with calibration error of 2.5%.

Keep the other two as the sensitivity analysis. They answer the obvious reviewer question — what happens if you drop the period where the microstructure features are missing, and what happens if you keep it — and they show the answer is small at 1 session and material at 10.

Say what is still missing. 154 sessions from 2022-07-28 to 2023-03-30 are absent from the archive, and the paper should name them rather than let a reader infer continuous coverage from 2018 to 2026.

Both the models and the stand-in caveat are unchanged. CatBoost, LightGBM, XGBoost, the Transformer and the TFT could not be installed in this environment; all three runs use scikit-learn stand-ins. If the native lite-panel run is repeated, it should start from the supplemented floorsheet and apply the same rule.

Reproduction

The recovered files are parsed by a new stage, `s02b_supplement.py`, which applies the same validation, de-duplication and daily aggregation as stage 02 and merges the result into `daily_floorsheet.csv.gz`, replacing a bar only when the recovered archive contains that symbol-session. Its report is `stage02b_supplement_report.json`.

`python s02b_supplement.py --src <dir>` then `NEPSE_EXCLUDE_GAP=1 FAMILY=sklearn START=3 bash run_pipeline.sh`. Stages 1–2 (the 91-million-trade ingest) are not rerun; the cleaned daily floorsheet is reused and supplemented. The rebuild was verified against the earlier run before any change: panel rows 405,491, eligible 360,943, 362 symbols, 1,855 sessions and all four label counts reproduced exactly.